Current Formulation of `src/model.py`: Head-wise Attention Residual

Experiment log, branch `attention-experiments`

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Summary

With `attn_residual=True` (the default in `default_hparams`), every attention head attends over $V_h + X_h$, where X_h is that head's slice of the raw residual stream. The attention output then *replaces* the residual stream instead of being added to it. The MLP sublayer is unchanged. Terms that differ from standard GPT-2 are highlighted.

Notation

Row-vector convention: each token is a row and weights multiply on the right (`x @ W`), which matches `conv1d`.

Symbol	Meaning	GPT-2 small
T	sequence length	–
D	model width (`n_embd`)	768
H	number of heads (`n_head`)	12
$d_h = D/H$	per-head width	64
L	number of layers (`n_layer`)	12
V	vocabulary size	50257

1 Embedding (`model`)

$$X^{(0)} = W_{te}[\text{tokens}] + W_{pe}[\text{positions}] \in \mathbb{R}^{T\times D}$$

2 Transformer block $\ell = 0, \ldots, L-1$ (`block`, `attn`)

Write $X = X^{(\ell)}$.

LayerNorm (`ln_1`).

$$\tilde{X} = \mathrm{LN}_1(X) = \frac{X-\mu}{\sqrt{\sigma^2+\epsilon}} \odot g_1 + b_1$$

Q, K, V (`c_attn`). The fused weight is $W_{QKV} \in \mathbb{R}^{D\times 3D}$:

$$[\,Q \mid K \mid V\,] = \tilde{X}\, W_{QKV} + b_{QKV}, \qquad Q_h, K_h, V_h \in \mathbb{R}^{T\times d_h}.$$

Per-head residual slice (new). This is taken from the *pre-LayerNorm* stream:

$$X_h = X[:,\ hd_h : (h+1)d_h] \in \mathbb{R}^{T \times d_h}$$

Attention pattern (unchanged).

$$S_h = \frac{Q_h K_h^\top}{\sqrt{d_h}}, \qquad S_h[i,j] \leftarrow -10^{10} \text{ for } j > i, \qquad A_h = \mathrm{softmax}_{\mathrm{row}}(S_h).$$

Head output (modified).

$$O_h = A_h\ (V_h + X_h) \in \mathbb{R}^{T \times d_h} \qquad \text{(standard: } O_h = A_h V_h\text{)}$$

Output projection W_O (`attn/c_proj`). W_O is the variable `model/hℓ/attn/c_proj/w`, with shape $[1, D, D]$ and used as $D \times D$:

$$\mathrm{Attn}(X) = [\, O_1 \mid O_2 \mid \cdots \mid O_H \,]\, W_O + b_O.$$

Attention residual (modified).

$$X' = \mathrm{Attn}(X) \qquad \text{(standard: } X' = X + \mathrm{Attn}(X)\text{)}$$

MLP with identity residual (unchanged). $W_{fc} \in \mathbb{R}^{D \times 4D}$ and $W_{proj} \in \mathbb{R}^{4D \times D}$:

$$X^{(\ell+1)} = X' + \mathrm{GELU}\big(\mathrm{LN}_2(X')\, W_{fc} + b_{fc}\big)\, W_{proj} + b_{proj}.$$

3 Output (`ln_f`, tied embedding)

$$\mathrm{logits} = \mathrm{LN}_f\big(X^{(L)}\big)\, W_{te}^\top \in \mathbb{R}^{T \times V}$$

4 Expanded attention sublayer

Let $W_O^{(h)} = W_O[hd_h : (h{+}1)d_h,\ :] \in \mathbb{R}^{d_h \times D}$ be the rows of W_O that belong to head h. Concatenation followed by W_O is a sum over heads, so

$$X' \ = \ \underbrace{\sum_h A_h V_h\, W_O^{(h)}}_{\text{standard attention}} \ + \ \underbrace{\sum_h A_h X_h\, W_O^{(h)}}_{\text{replaces the identity skip } X} \ + b_O.$$

5 Implications

- **The residual is mixed across positions.** Token i no longer keeps its own state X_i. It receives $\sum_j A_h[i,j]\, X_{h,j}$, a mixture of earlier tokens' residual slices weighted by that head's attention pattern.

- **The residual goes through W_O.** The skip path becomes $\sum_h A_h X_h W_O^{(h)}$ instead of the identity. The pretrained W_O was trained to project value vectors, not to reconstruct X, so the stream is distorted in every layer.
- **Pre-LN versus post-LN inputs.** V_h comes from $\mathrm{LN}_1(X)$, but X_h comes from the raw X. Their scales can differ substantially, especially in deep layers where the norm of the residual stream grows.
- **The effect compounds over depth.** Only the MLP sublayer keeps a true identity path, and the attention replacement applies in all $L = 12$ layers.
- **No KV cache.** X_h is needed at every source position, so `model()` asserts `past is None`.

Code references

`src/model.py:25`	`attn_residual=True` flag
`src/model.py:113{115`	$O_h = A_h(V_h + X_h)$
`src/model.py:126{131`	$X_h =$ `split_heads(residual_x)`
`src/model.py:134{135`	`merge_heads`, then W_O (`c_proj`)
`src/model.py:150{154`	$X' = \mathrm{Attn}(X)$ replaces $X + \mathrm{Attn}(X)$